

Checklists — Part 2

Safety Inspection / Audit, Mast & Sub-Structure Inspection and Contractor Safety Inspection

⚠ FOR UNDERSTANDING PURPOSES ONLY

Prasanaa Kartick T — Personal Learning Project

Flutter / Dart

Inspection / Audit

Mast & Substructure

Contractor audit

PDF

1. Safety Inspection / Audit Checklist

A full rig audit, in **seventeen lettered sections**. Each line is a statement to confirm, with a status that cycles on a single tap, and the results export as an A4 PDF.

Section	Examples of what is checked
A. Site header, documents and records	Company, rig and manager details, licences and permits posted, incident report forms, contingency plan for critical H2S wells, well-control certificates, BOP drill and function-test records, H2S and induction training, first aid certificates, drill-line ton-mileage records, emergency numbers posted
B. Location and general housekeeping	Site signs, overhead lines, vehicle position, drainage, trip hazards, walkways, spill kits, waste segregation, smoking areas
C. PPE	Hard hats, safety shoes, eye and hearing protection, gloves, fire-retardant clothing, harnesses, respiratory protection
D. Fire prevention and protection	Extinguishers, hose reels and hydrants, fire and gas detection, foam systems, hot work permits, flammable storage, signage, alarms
E. Electrical safety	Wiring, explosion-proof lighting, intrinsically safe equipment in classified zones, tagged portable tools
F to Q	Chemicals and hazardous materials, work permits, substructure and cellar, BOP, accumulator and choke manifold, rig floor and doghouse, derrick and travelling assembly, drawworks and catheads, pipe rack, mud pumps, mud tanks and solids control, generator and engine room, storage areas and fuel tanks

2. Mast & Sub-Structure Inspection

Unlike the other checklists, this one is an **illustrated reference guide**, with no form to fill in. It explains how the mast, substructure and other drilling equipment should be inspected and how often. Its goals are shown at the top: **detect damage early, prevent failure, ensure safe and reliable operations**.

The four inspection levels

Level	Name	Who and how often
I	Primary / routine	Rig personnel, visually, during operations. Daily and weekly
II	Detailed	Designated experienced personnel, during rig-up and periodically
III	Advanced (NDT of critical areas)	Qualified personnel, every two years , with partial disassembly where needed
IV	Comprehensive (full NDT)	OEM or a professional engineer, every ten years , with disassembly in a suitable facility

Each level has its own panel with a large picture, a description, the items to check, the frequency, and the actions required. Level I looks for lost coating, impact damage, corrosion, cracks, loose nuts and bolts and similar visible defects. Level IV adds ultrasonic thickness tests, internal camera inspection and magnetic particle inspection on welds.

The rest of the guide

- **Key reminders:** follow the inspection frequency strictly, keep records for every inspection and remove unsafe equipment from service immediately.
- **Important notes:** schedules should consider corrosive environments, temperature and the owner's own schedule.
- **Frequency table:** a recommended inspection frequency, for drilling and workover equipment, of daily or weekly, six-monthly, yearly and five-yearly checks. It covers crown block sheaves, hooks, travelling blocks, elevators, slips, tongs, the kelly, rotary swivels and hose, dead-line anchors, motion compensators, the drawworks and the rotary table.
- **Equipment pages:** for example the deadline anchor, with what to check on its wire-line clamp inserts, drum grooves and foundation bolts.

3. Contractor Safety Inspection / Audit

An audit checklist for contractor work, in **nineteen sections** that follow the type of work. It uses the N/A → Yes → No cycle, includes a remarks and actions box, and exports as a PDF.

Group	Sections
General	Personal protective equipment · Housekeeping · Permits · General · Emergency procedures · Welfare facilities
Hot and heavy work	Cutting, welding and grinding · Sand and shot blasting · Material handling
Access and environment	Excavation · Working at heights, scaffolding and ladders · Confined space · Electrical safety
Civil work	Road work · Form work and reinforcement · Concreting · Demolishing
Special hazards	Radiography · Units with hydrocarbons

- **Header:** job name, location, inspected by, contractor or contractors, date of audit and frequency.
- **Custom rows** can be added to any section.
- **Remarks and actions:** a free-text box for overall remarks and corrective actions.
- **Sign-off:** signature of the auditor and of the contractor's representative on the PDF.

4. Why So Many Checklists

One idea, applied many ways

All seven checklists use the same building blocks: a row, a status that cycles on a tap, optional custom rows, a header, a PDF. That is why they look and behave alike, and why a new checklist can be added with little extra work.